LAB 08

Execution Stage: an Order Manager that Trades the Paper Account in Any Session

Item	Detail
Topic	Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca
Pipeline stage	Execution
TSC reference	A3, K5
Duration	30 min
Folder	labs/Lab08-paper-orders/
Files	Lab08-paper-orders/execution_stage.py, pipeline/execution.py, pipeline/clock.py, common/broker.py, Lab08-paper-orders/solution/execution.py
Simulator page	http://localhost:8000/#/trading?tab=ticket
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

The pipeline now produces target positions. The desk needs an order manager that turns 'target' versus 'actual' into paper orders safely: the right order type for the current US session (the Singapore class runs while New York is in its overnight and pre-market sessions), prices off the live quote, no duplicate orders if a job runs twice, and a printed order lifecycle the trader can match in the Alpaca dashboard.

Objective

Build pipeline/execution.py: a session-aware, idempotent order manager that sizes orders from target weights, sends market, extended-hours limit, GTC-queued or bracket PAPER orders, polls their lifecycle, cancels protective legs before exits, and falls back safely when an order is rejected.

Before You Start

- Part A: start the Trading Simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000 > Course Labs > Lab 08. Install steps: Learner Guide, 'Download, Install and Start the Trading Simulator Software'.
- Part B: open a terminal in the labs folder, activate your virtual environment and open your AI coding assistant there; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

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Part A - In the Trading Simulator

Start the simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000. In the sidebar open Course Labs > Lab 08: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

A1. Market order in the regular session

Paper Trading > Order ticket: BUY 10 SPY, type market, Session (sim) = regular. Send.

Open: http://localhost:8000/#/trading?tab=ticket (Paper Trading > Order ticket)

Check: Result ACCEPTED and status filled; every risk check is listed.

A2. Extended-hours limit order (Singapore class hours)

Same ticket, type auto, Session (sim) = overnight. The app converts it to a LIMIT order, time in force DAY, extended_hours=True, priced off the quote ± 25 bps.

Open: http://localhost:8000/#/trading?tab=ticket

Check: The order shows 'limit day ext-hrs'. A market order in the overnight session is blocked.

A3. Bracket order

BUY 5 SPY, regular session, with a take-profit and stop-loss price. Then open the Orders tab and expand the status to see the lifecycle and the two exit legs (held until the parent fills, then one-cancels-other).

Open: http://localhost:8000/#/trading?tab=orders

Check: Two legs appear: a limit (take-profit) and a stop (stop-loss).

A4. Queue while closed and cancel

Session = closed: the order becomes a GTC limit that queues for the next session. Cancel it from the Orders tab.

Open: http://localhost:8000/#/trading?tab=orders

Command line: With Alpaca keys: TradingClient(api_key, secret_key, paper=True).submit_order(LimitOrderRequest(symbol='SPY', qty=1, side=OrderSide.BUY, time_in_force=TimeInForce.DAY, limit_price=..., extended_hours=True))

Check: Status moves accepted -> canceled.

Expected in the app: A blotter with filled, working, queued and cancelled orders and the risk checks for each.

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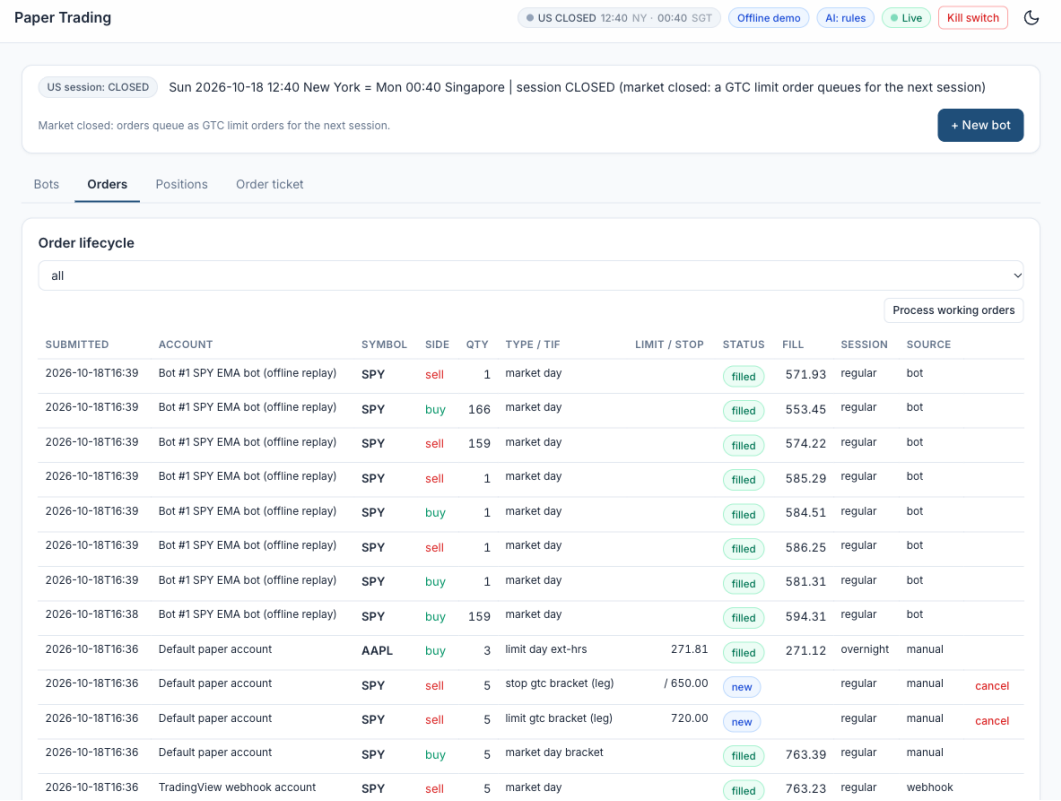


Figure: Lab 08 in the Trading Simulator (Lab08_paper_orders.png).

Part B - In Code with Your AI Assistant

The same pipeline stage in code. AI writes the code; you own the algorithm: paste each prompt into your AI coding assistant in the labs folder (the finished reference code is already in labs/Lab08-paper-orders/), run it, and check the result against Part A. The Practical Performance assessment uses this code path.

Step 1: Learn the order rules for each session (K5)

Open pipeline/clock.py and read RULES. Alpaca 24/5 trading: overnight 20:00 Sunday to 04:00 Friday (New York), pre-market 04:00-09:30, regular 09:30-16:00, after-hours 16:00-20:00. Outside the regular session only LIMIT orders with time_in_force DAY or GTC and extended_hours=True are accepted.

AI prompt - paste into your AI coding assistant
Using labs/pipeline/clock.py, tell me which US session it is now in New York, what that is in Singapore time, and which order types, time_in_force values and extended_hours flag Alpaca will accept right now. Then explain why a market order is not allowed overnight. Do not change any file.

Check: You can name the session for a 09:00-18:30 SGT class (OVERNIGHT until 16:00 SGT in US summer time, then PRE) and state 'limit only, extended_hours=True'.

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Step 2: Build the planning logic: target vs actual (A3)

Ask your assistant for the OrderPlan dataclass and the planning methods.

AI prompt - paste into your AI coding assistant
Create labs/pipeline/execution.py with a dataclass OrderPlan(symbol, side, qty, current_qty, pending_qty, target_qty, price, reason) and a class OrderManager(broker, cfg, session=None). status() returns pipeline.clock.market_status (the simulator's replay clock if set; the real Alpaca clock online; session=... overrides it offline) and tells the simulator whether the market is open. positions() -> {symbol: qty}; pending() -> signed shares on open orders (+buy, -sell) via GetOrdersRequest(status=OPEN); quote(symbol, session) -> bid/ask/last: online StockLatestQuoteRequest (feed OVERNIGHT first in the overnight session, then IEX), falling back to the latest trade; offline the simulator's last price +/- 0.5 bp. plan(targets, session, units='weight'|'shares'): target_qty = floor(weight x equity / price) (or the share count), delta = target - held - pending, buy if delta > 0, sell (long only, never more than held) if delta < 0, otherwise print 'at target' and send nothing. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: OrderManager(get_broker(state_file='exec_sim.json', reset=True), load_config()).plan({'SPY': 0.30}, 'regular') returns one buy plan for 44 shares.

Step 3: Build session-aware, idempotent order submission (A3, K5)

Ask your assistant for client ids, the order request builder and submit().

AI prompt - paste into your AI coding assistant
In labs/pipeline/execution.py add client_id(plan, tag) = 'lcc-<tag>-<SYMBOL>-<side>-<qty>' (alphanumerics and dashes, max 48 characters; tag = decision date) so the same decision always gets the same id. request(plan, order_type, status, cid, quote, take_profit=0.05, stop_loss=0.03) builds an alpaca-py request: regular + market -> MarketOrderRequest DAY; regular + bracket (buys) -> MarketOrderRequest GTC with OrderClass.BRACKET, TakeProfitRequest and StopLossRequest; closed -> LimitOrderRequest GTC (queues for the next session); pre/after/overnight -> LimitOrderRequest DAY with extended_hours=True. Limit price = ask x (1 + live_paper.limit_buffer_bps) for buys, bid x (1 - buffer) for sells, rounded to 2 decimals. submit(plan, status, tag, order_type='auto'): first broker.get_order_by_client_id(cid) - if it exists print 'DUPLICATE ... not sent again' and return it; print the order (type, TIF, extended_hours, limit, quote and source, cid); send it with broker.submit_order. The broker comes from common.broker.get_broker (Alpaca TradingClient with paper=True, or PaperBrokerSim) - never create a live client. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: Submitting the same plan twice with the same tag prints DUPLICATE the second time.

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Step 4: Add polling, protective-order handling and safe fallbacks (A3, K5)

Ask your assistant for poll(), cancel_protective(), the rejection fallback, the crypto demo and the command.

AI prompt - paste into your AI coding assistant
In labs/pipeline/execution.py add: poll(order, seconds) that re-reads the order every 2 seconds and prints the lifecycle (for example 'new -> filled') and the final order line, or '(still working - it stays on the book)'; cancel_protective(symbol) that cancels open SELL orders and bracket exit legs (GetOrdersRequest nested=True) before an exit, because they reserve the shares; in submit(), if an extended-hours order is rejected, print why (asset or account not enabled for overnight trading, price band, session change) and fall back to a GTC limit order with client id + '-q' that queues for the next session; rebalance(targets, tag, order_type, poll_seconds, units) = status -> plan -> submit -> poll; crypto_demo(broker, cfg) - ONLINE ONLY, clearly labelled DEMO: buy about USD 25 of BTC/USD with a GTC limit at ask + buffer, poll, then sell it back - only to show the fill loop working while US stocks are closed. run(...) and main(argv) for `python -m pipeline execute [--targets SPY=0.30 AAPL=0.10] [--order-type auto|market|limit|bracket] [--session ...] [--live-paper] [--crypto-demo] [--flatten] [--tag] [--reset]`: print [1] the clock and allowed orders, [2] the account, [3] plan -> orders, [4] an idempotency re-run of the same decision plus a deliberate duplicate submission, [5] positions and open orders (with bracket legs), the Alpaca dashboard address when online, and a RESULT line. With --live-paper or online, cap each weight at live_paper.max_position_pct. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline execute --help lists the eight options.

Step 5: Run the order manager in the current session (A3, K5)

Run: python -m pipeline execute --reset (same as: python3 Lab08-paper-orders/execution_stage.py --reset). Offline the session comes from the real clock, so the result depends on when you run it.

Check: Weekend (CLOSED): BUY 44 SPY at LIMIT GTC 678.36 and BUY 36 AAPL at 271.81, both 'accepted' (queued); the re-run prints 'at target (44 shares; held 0, pending +44)' and 'DUPLICATE lcc-<date>-SPY-buy-44'. Weekday Singapore class (OVERNIGHT or PRE): LIMIT DAY extended_hours orders, filled at about 676.63 and 271.12.

Step 6: Compare the sessions with the teaching override (K5)

Run: python -m pipeline execute --reset --session regular, then --session overnight, then --session closed.

Check: regular: MARKET DAY, filled at 676.70 (SPY) and 271.15 (AAPL), equity USD 99,996.05; overnight: LIMIT DAY extended_hours limit 678.36 / 271.81, filled at 676.63 / 271.12, equity USD 100,000.00; closed: LIMIT GTC, status accepted, 2 open orders. Each run ends with 'no new orders' and one DUPLICATE line.

Step 7: Bracket orders and a clean exit (A3, K5)

Run: python -m pipeline execute --reset --session regular --order-type bracket. Then flatten: python -m pipeline execute --flatten --session regular.

Check: Bracket legs: SPY take-profit 710.47 / stop 656.34, AAPL 284.68 / 262.99 (status new). Flatten prints 'cancelled 2 protective exit order(s)' for each symbol, sells 44 SPY at 676.57 and 36 AAPL at 271.09, equity USD 99,992.09, 0 open orders.

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Step 8: Trade LIVE on the Alpaca paper account during class (A3, K5)

With paper keys in .env run: python -m pipeline execute --live-paper --targets SPY=0.05. Watch the clock line, the quote, the extended-hours limit order and its lifecycle for up to 30 seconds. Then open the Alpaca paper dashboard (app.alpaca.markets > Paper > Orders and Positions) and find the same client order id.

Check: Clock shows OVERNIGHT or PRE; one LIMIT DAY extended_hours buy priced at ask + 25 bps; lifecycle 'new -> filled' (or still working); the order and position appear in the dashboard. If Alpaca rejects it, the stage explains why and queues a GTC limit order instead.

Step 9: Optional - watch a fill right now with the crypto demo (K5)

If US stocks are closed (weekend) or your extended-hours order is still working, run: python -m pipeline execute --live-paper --targets SPY=0.05 --crypto-demo. This is a DEMO of the fill loop only; the course and the pipeline remain stock-focused.

Check: A BTC/USD buy of about USD 25 and the matching sell, each GTC limit, filled within seconds and visible in the dashboard.

Step 10: Flatten the live paper account (A3)

Run: python -m pipeline execute --live-paper --flatten, then refresh the dashboard. Offline, run the same command without --live-paper.

Check: Protective orders are cancelled, sell orders are sent (extended-hours limits outside regular hours) and the positions list empties once they fill.

Expected Output

python -m pipeline execute --reset --session regular (offline): BUY 44 SPY and BUY 36 AAPL as MARKET DAY orders, filled at 676.70 and 271.15, equity USD 99,996.05; the idempotency re-run sends no new order and a duplicate submission is skipped; --session overnight sends LIMIT DAY extended_hours orders (filled 676.63 / 271.12); --session closed queues LIMIT GTC orders (accepted); bracket legs SPY 710.47 / 656.34; --flatten cancels the protective legs and exits (equity USD 99,992.09). With --live-paper the same flow runs on the Alpaca paper account in any session.

Guide: Paper Trading on Alpaca, Step by Step - Manual and Automated

First place one order by hand to see the order lifecycle, then let the pipeline place orders automatically (Labs 8 and 16).

1. Place a manual paper order

In the Alpaca dashboard select the Paper Trading account (upper left), search the symbol (for example SPY) and open its page. Set the side, quantity, order type and time in force, then press Buy.

2. Follow the order lifecycle

Open the Orders view and watch the status move from new / accepted to filled (or cancelled). Open Positions to see the holding, cost basis and unrealised P&L.

3. Know the trading hours from Singapore

The US regular session runs 21:30-04:00 Singapore time (22:30-05:00 in the northern winter). Outside it, Alpaca's 24/5 trading accepts only LIMIT orders with extended hours enabled (time in force DAY or GTC), at lower liquidity; a market order placed while the market is closed waits for the next session.

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4. Let the pipeline place the orders

In the labs folder run: python -m pipeline execute (Lab 8). The order manager reads the Alpaca clock, compares the target position with the actual one, submits the paper order, polls its status and prints the fill.

5. Run the strategy automatically

Run: python -m pipeline run --loop --interval 60 --live-paper (Lab 16). Each cycle fetches bars, computes the signal, runs the risk checks, places or skips the paper order and writes a row to the trade log.

6. Check the results in the dashboard

Refresh the Alpaca dashboard: the bot's orders appear under Orders and its holdings under Positions, matching labs/_outputs/trade_log.csv.

7. Stop it safely

Run: python -m pipeline kill --on. The next cycle halts before any order is sent. Cancel any working orders and, if you wish, close positions from the dashboard.

From Singapore the US regular session is 21:30-04:00 (22:30-05:00 in northern winter); class-time orders use Alpaca's 24/5 extended-hours limit orders.

Debrief

Order types are tools for price, timing and risk, and the trading session decides which ones the broker accepts (K5). An order manager that reads the clock, prices limits off the quote, makes every order idempotent and prints its lifecycle is how an automated program knows exactly what happened at the broker (A3) - in New York's regular session or in its overnight session during a Singapore workday.

Troubleshooting

Symptom	What to do
Extended-hours order rejected online	The asset or account may not be enabled for overnight trading or the limit is outside the price band; the stage falls back to a GTC limit that queues for the next session.
Order stays 'new' or 'accepted'	Normal outside regular hours or when the market is closed; it remains working - check the Orders page in the dashboard.
insufficient qty available	A stop or bracket leg reserves the shares; cancel_protective() must run before any sell.
insufficient buying power	Run with --reset offline, or lower the target weights.

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